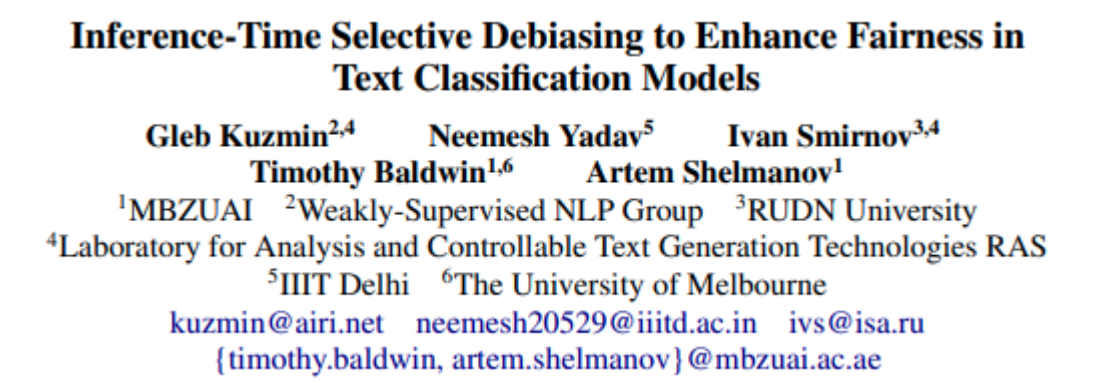

Inference-Time Selective Debiasing to Enhance Fairness in Text Classification Models

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Discussing Inference-Time Selective Debiasing to Enhance Fairness in Text Classification Models

Paper by Gleb Kuzmin, Neemesh Yadav, Ivan Smirnov, Timothy Baldwin, Artem Shelmanov

Presentation by John Naughton 2025

Introduction

Hello, my name is John Kieran Naughton, I am in the Masters of Data Science online program here at UCSD.

For my Natural Language Processing 258R video assignment, I will be discussing the main concepts, findings, and implications of the 2025 paper “Inference-Time Selective Debiasing to Enhance Fairness in Text Classification Models” by Gleb Kuzmin, Neemesh Yadav, Ivan Smirnov, Timothy Baldwin, and Artem Shelmanov. The paper was published in April 2025 by the Association for Computational Linguistics

Summary of Inference-Time Selective Debiasing

- This paper proposes **selective debiasing**, an inference-time method to enhance fairness and performance in text classification without retraining. It identifies biased predictions that have large uncertainty values using a Kullback-Leibler divergence-based bias quantification that has the potential to outperform standard uncertainty measures. Instead of discarding these outputs and overfitting, the method applies LEACE, a post-processing technique, to remove bias from these selected predictions. The experiments conducted show selective debiasing significantly closes the performance gap between post-processing and more resource intensive retraining-dependent debiasing methods.

Bias Mitigation technique: Inference-Time Selective Debiasing

- Traditional debiasing techniques assume access to the training data and the ability to either retrain the model from scratch or reweight the training instances.

- Existing selective classification enhances the reliability of ML-based applications by abstaining from unreliable predictions with high uncertainty. This paper proposes an inference-time safety mechanism instead, where they create a scoring criterion that selects the most unreliable and biased predictions, and then applies the traditional post-training (LEACE in this case) to only the selected predictions.

Background on debiasing techniques

Debiasing techniques are categorized into at-training, pre-processing, and post-processing. At-training (like Adv, DAdv) and pre-processing (like BTEO, BTJ) methods require retraining and full data access. Post-processing (INLP, LEACE) modifies predictions without retraining and can be selective. The paper introduces a method combining post-processing's flexibility with better performance.

Kullback-Leibler Divergence

To identify these "problematic" predictions, the paper introduces a novel bias quantification approach based on KL divergence.

$$B^i_{KL} = \sum_{c \in C} p^c_i \log\left(\frac{p^c_i}{\bar{p}_i{}^c}\right)$$

This method measures the difference between the original predicted probability distribution and the distribution after debiasing. A greater KL divergence indicates more information about the protected attribute was removed, suggesting a higher initial bias. Experiments show this KL-based approach is more effective than standard uncertainty quantification methods for selecting instances to debias. This selective application of debiasing helps to bridge the performance gap often seen between post-processing methods and more resource-intensive at-training or pre-processing techniques.

Debias technique: LEACE

LEAst-squares Concept Erasure (LEACE) is a post-processing debiasing technique that precisely removes sensitive "concepts" from a model's internal data representations without needing to retrain the entire model. It works by first standardizing the variance across all dimensions of the representations (aka whitening). Then it calculates a specific projection that targets and eliminates the linear information within these representations that is correlated with the protected attribute. Finally, it reverses the initial standardization and applies this calculated transformation to the original representations. This results in a direct, non-iterative calculation with new, debiased representations all while making minimal changes to the representations themselves.

Experiment Design

The experiments used two English text classification datasets with explicit protected attributes: MOJI for sentiment analysis (binary class: "happy“ or "sad"; binary protected attribute: African American English (AAE) vs. Standard American English (SAE) ethnicity of the author) and BIOS-2 for occupation classification (binary gender as the protected attribute). Both balanced and imbalanced versions of these datasets were tested. The models used were BERT for BIOS-2 and BERTweet for MOJI, each supplemented with a three-layer MLP classification head. The study compared the proposed selective debiasing (using LEACE with KL-divergence for bias quantification, benchmarked against Softmax Response) against a suite of baselines including at-training methods (Adv, DAdv), pre-processing methods (BTEO, BTJ), and other post-processing methods (INLP, full LEACE). Notably the at-training and pre-processing methods required full dataset retraining while the post-processing techniques, including our selective debiasing, were trained using only 20% of the data.

Four metrics are used: Fairness, Accuracy, DTO and FF-score

$$\text{DTO} = \sqrt{(1-\text{Accuracy})^2 + (1-\text{Fairness})^2}$$

The study employs several metrics to comprehensively evaluate model performance and fairness. Predictive performance is measured by accuracy. Fairness is specifically assessed using the equal opportunity criterion, a widely recognized group fairness definition. Additionally, two aggregated metrics are used to capture both accuracy and fairness: Distance to Optimal (DTO), which quantifies the Euclidean distance from ideal accuracy and fairness, and Fairness F-score (FF-score), a harmonic mean-like measure that provides a smoothed minimum of accuracy and fairness.

$$\text{FF-score} = \frac{2 \cdot \text{Accuracy} \cdot \text{Fairness}}{\text{Accuracy} + \text{Fairness}}$$

Debiasing method type		No debiasing	At-training		Pre-processing		Post-processing & Selective					
Dataset	**Metric**	**Standard**	**Adv**	**DAdv**	**BTEO**	**BTJ**	**LEACE-last**	**LEACE-last+SR, opt. perc.**	**LEACE-last+KL, opt. perc.**	**LEACE-cls**	**LEACE-cls+SR, opt. perc.**	**LEACE-cls+KL, opt. perc.**
MOJI imbalanced	Fairness ↑	$61.8_{\pm 0.7}$	$73.7_{\pm 0.6}$	$73.4_{\pm 0.4}$	$\mathbf{75.2}_{\pm 0.6}$	$74.8_{\pm 0.6}$	$\mathbf{75.8}_{\pm 2.6}$	$68.6_{\pm 1.4}$	$75.7_{\pm 0.8}$	$75.2_{\pm 3.0}$	$68.4_{\pm 1.1}$	$\underline{\mathbf{77.2}}_{\pm 0.7}$
	Accuracy ↑	$\underline{\mathbf{79.1}}_{\pm 0.7}$	$72.0_{\pm 0.7}$	$72.4_{\pm 0.5}$	$73.6_{\pm 0.6}$	$73.2_{\pm 0.4}$	$68.3_{\pm 2.6}$	$\mathbf{77.6}_{\pm 0.9}$	$72.7_{\pm 1.2}$	$66.8_{\pm 3.0}$	$\mathbf{77.6}_{\pm 1.0}$	$71.8_{\pm 1.2}$
	DTO ↓	$43.6_{\pm 0.6}$	$38.4_{\pm 0.5}$	$38.3_{\pm 0.4}$	$\underline{\mathbf{36.2}}_{\pm 0.1}$	$36.7_{\pm 0.4}$	$39.9_{\pm 3.6}$	$38.6_{\pm 0.7}$	$\mathbf{36.6}_{\pm 0.8}$	$41.4_{\pm 4.1}$	$38.8_{\pm 0.6}$	$\underline{\mathbf{36.2}}_{\pm 1.2}$
	FF-score ↑	$69.4_{\pm 0.4}$	$72.8_{\pm 0.4}$	$72.9_{\pm 0.3}$	$\underline{\mathbf{74.4}}_{\pm 0.1}$	$\mathbf{74.0}_{\pm 0.3}$	$71.8_{\pm 2.6}$	$72.8_{\pm 0.5}$	$\mathbf{74.1}_{\pm 0.3}$	$70.8_{\pm 3.0}$	$72.7_{\pm 0.4}$	$\underline{\mathbf{74.4}}_{\pm 0.8}$
MOJI balanced	Fairness ↑	$69.5_{\pm 0.2}$	$83.8_{\pm 0.8}$	$84.7_{\pm 1.5}$	$85.5_{\pm 0.5}$	$\mathbf{85.6}_{\pm 0.6}$	$79.7_{\pm 3.9}$	$77.1_{\pm 0.9}$	$\mathbf{86.6}_{\pm 0.8}$	$77.6_{\pm 4.2}$	$77.0_{\pm 0.8}$	$\underline{\mathbf{87.5}}_{\pm 0.8}$
	Accuracy ↑	$71.9_{\pm 0.4}$	$74.0_{\pm 0.4}$	$74.1_{\pm 0.6}$	$\underline{\mathbf{74.8}}_{\pm 0.3}$	$74.5_{\pm 0.4}$	$73.6_{\pm 0.8}$	$\mathbf{74.0}_{\pm 0.3}$	$74.0_{\pm 0.2}$	$73.0_{\pm 1.2}$	$\mathbf{74.0}_{\pm 0.4}$	$73.7_{\pm 0.5}$
	DTO ↓	$41.5_{\pm 0.4}$	$30.7_{\pm 0.7}$	$30.1_{\pm 0.7}$	$\underline{\mathbf{29.0}}_{\pm 0.1}$	$29.3_{\pm 0.4}$	$33.4_{\pm 3.0}$	$34.7_{\pm 0.7}$	$\mathbf{29.3}_{\pm 0.3}$	$35.2_{\pm 3.7}$	$34.7_{\pm 0.6}$	$\mathbf{29.1}_{\pm 0.6}$
	FF-score ↑	$70.7_{\pm 0.3}$	$78.6_{\pm 0.5}$	$79.1_{\pm 0.6}$	$\mathbf{79.8}_{\pm 0.1}$	$79.6_{\pm 0.3}$	$76.5_{\pm 2.2}$	$75.5_{\pm 0.5}$	$\mathbf{79.8}_{\pm 0.2}$	$75.2_{\pm 3.6}$	$75.5_{\pm 0.4}$	$\underline{\mathbf{80.0}}_{\pm 0.4}$
BIOS-2 imbalanced	Fairness ↑	$90.4_{\pm 0.8}$	$\underline{\mathbf{97.2}}_{\pm 0.8}$	$96.4_{\pm 0.4}$	$95.8_{\pm 1.0}$	$96.6_{\pm 0.8}$	$92.8_{\pm 0.3}$	$93.0_{\pm 2.3}$	$94.5_{\pm 4.4}$	$77.3_{\pm 6.5}$	$94.8_{\pm 2.3}$	$\mathbf{96.7}_{\pm 0.9}$
	Accuracy ↑	$\underline{\mathbf{96.7}}_{\pm 0.1}$	$94.8_{\pm 0.4}$	$95.0_{\pm 0.3}$	$95.2_{\pm 0.3}$	$95.0_{\pm 0.5}$	$60.5_{\pm 3.6}$	$\mathbf{94.6}_{\pm 0.2}$	$92.0_{\pm 0.4}$	$64.0_{\pm 5.5}$	$\mathbf{94.6}_{\pm 0.1}$	$93.2_{\pm 0.3}$
	DTO ↓	$10.1_{\pm 0.7}$	$\underline{\mathbf{5.9}}_{\pm 0.2}$	$6.2_{\pm 0.2}$	$6.5_{\pm 0.6}$	$6.1_{\pm 0.3}$	$41.3_{\pm 2.1}$	$\mathbf{9.0}_{\pm 1.5}$	$10.3_{\pm 2.1}$	$43.4_{\pm 2.4}$	$7.7_{\pm 1.3}$	$\mathbf{7.6}_{\pm 0.5}$
	FF-score ↑	$93.5_{\pm 0.4}$	$\underline{\mathbf{96.0}}_{\pm 0.2}$	$95.7_{\pm 0.1}$	$95.5_{\pm 0.4}$	$95.8_{\pm 0.2}$	$72.8_{\pm 2.3}$	$\mathbf{93.8}_{\pm 1.2}$	$93.2_{\pm 2.3}$	$69.6_{\pm 1.7}$	$94.7_{\pm 1.2}$	$\mathbf{94.9}_{\pm 0.4}$
BIOS-2 balanced	Fairness ↑	$89.7_{\pm 0.6}$	$97.8_{\pm 0.8}$	$\underline{\mathbf{98.0}}_{\pm 0.8}$	$95.9_{\pm 0.8}$	$96.4_{\pm 0.3}$	$90.6_{\pm 0.8}$	$93.7_{\pm 2.6}$	$94.6_{\pm 4.2}$	$74.8_{\pm 9.2}$	$96.6_{\pm 1.8}$	$\mathbf{97.5}_{\pm 0.9}$
	Accuracy ↑	$92.4_{\pm 0.3}$	$91.9_{\pm 0.6}$	$91.9_{\pm 1.5}$	$92.6_{\pm 0.5}$	$\underline{\mathbf{92.9}}_{\pm 0.6}$	$49.9_{\pm 0.4}$	$\mathbf{90.9}_{\pm 1.3}$	$89.3_{\pm 1.8}$	$63.8_{\pm 10.1}$	$\mathbf{91.9}_{\pm 0.7}$	$90.6_{\pm 1.3}$
	DTO ↓	$12.8_{\pm 0.6}$	$8.5_{\pm 0.4}$	$8.4_{\pm 1.4}$	$8.5_{\pm 0.2}$	$\underline{\mathbf{8.0}}_{\pm 0.6}$	$52.4_{\pm 6.0}$	$\mathbf{11.1}_{\pm 2.4}$	$12.4_{\pm 3.4}$	$46.0_{\pm 4.3}$	$\mathbf{8.9}_{\pm 1.4}$	$9.7_{\pm 1.5}$
	FF-score ↑	$91.1_{\pm 0.4}$	$94.7_{\pm 0.1}$	$\underline{\mathbf{94.9}}_{\pm 0.7}$	$94.2_{\pm 0.2}$	$94.6_{\pm 0.3}$	$63.0_{\pm 4.6}$	$\mathbf{92.3}_{\pm 1.9}$	$91.9_{\pm 2.3}$	$67.5_{\pm 3.6}$	$\mathbf{94.2}_{\pm 1.2}$	$93.9_{\pm 1.1}$

Table 1: Comparison of debiasing methods and selective debiasing. The best results in the group are in bold, and the best results overall are underlined. The results are averaged over 5 random seeds. The gray color corresponds to the results with p-value > 0.05 with respect to standard model.

Experiment Results

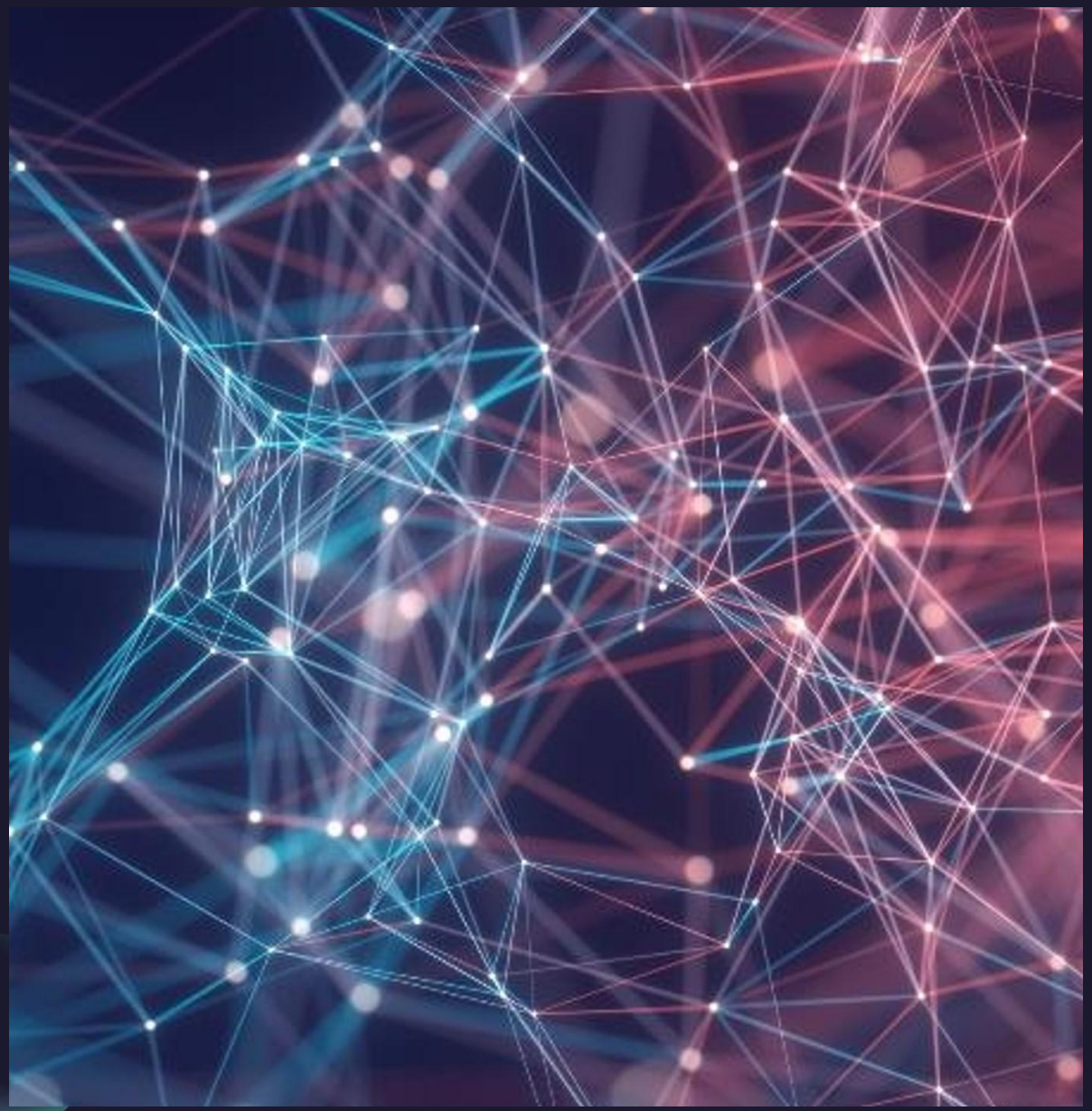
Unsurprisingly, methods that retrain models from scratch (at-training and pre-processing) generally yield the best performance. However selective debiasing that utilizes LEACE significantly boosts the results of inference-time debiasing, making it competitive with these more resource-intensive methods (especially when evaluating combined fairness and performance metrics like FF-score and DTO). On the MOJI-balanced dataset, selective debiasing with LEACE and KL selection even outperformed other methods. Furthermore, the paper's proposed KL-based bias quantification method consistently outperforms Softmax Response (SR) on MOJI datasets and performs comparably on BIOS-2. LEACE-cls generally showed better fairness and overall performance than LEACE-last. The findings consistently highlight that selective debiasing, particularly with KL-based selection, improves upon standard inference-time debiasing, offering a more efficient and effective solution for enhancing model fairness.

Paper Conclusions

The paper introduces selective debiasing, an innovative inference-time mechanism that significantly improves fairness and performance in text classification models. The selective debiasing is ideal for situations where retraining is impossible or full training data is unavailable. The method identifies biased predictions using a novel KL divergence-based bias quantification technique, which outperforms standard uncertainty measures. By enabling efficient debiasing during inference, this work “fills the gap” for efficient techniques that can be applied during the inference stage, and the authors hope to investigate deeper interactions between Uncertainty Quantification and various debiasing methods.

Thank you for listening

Have a nice summer!

John Kieran Naughton

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